

**KPR INSTITUTE OF ENGINEERING AND
TECHNOLOGY**

WEB TECHNOLOGIES ASSIGNMENT-1

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CLASS: III CSE D

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WEB TECHNOLOGIES ASSIGNMENT-1

1. **Scenario:** A newly established educational institution plans to launch its official website. Before

developing the website, the management wants to understand the fundamental concepts of the World Wide Web.

Task: Create a simple HTML webpage explaining the following concepts with suitable illustrations

or diagrams:

- Internet Overview
- Fundamental Computer Network Concepts
- URL and its Structure
- Domain Name
- Web Browsers and Web Servers
- Working Principle of a Website
- Difference between Client-side and Server-side Scripting

Program:

index.html:

```
<!DOCTYPE html>

<html>

<head>

<title>World Wide Web</title>

<link rel="stylesheet" href="style.css">

</head>

<body>

<header>

<h1>Fundamentals of World Wide Web</h1>

</header>

<div class="container">

<section>
```

<h2>Internet Overview</h2>

<p>The Internet is a global network that connects computers and allows them to share information.</p>

</section>

<section>

<h2>URL Structure</h2>

<p>A URL identifies a resource on the Internet.</p>

<div class="url">https://www.example.com/index.html</div>

<p>https → Protocol
example.com → Domain Name
index.html → Web Page</p>

</section>

<section>

<h2>Web Browser and Web Server</h2>

<div class="diagram">

<div>Browser</div>

→

<div>Internet</div>

→

<div>Server</div>

</div>

</section>

<section>

<h2>Client-side vs Server-side</h2>

<table>

<tr>

<th>Client-side</th>

<th>Server-side</th>

</tr>

<tr>

<td>Runs in browser</td>

```

<td>Runs on server</td>

</tr>

<tr>

<td>JavaScript</td>

<td>PHP, Python, Java</td>

</tr>

</table>

</section>

</div>

</body>

</html>

```

Style.css:

```

body {
font-family: Arial, sans-serif;
margin: 0;
background: #f2f6fa;
}
header {
background: #174a7c;
color: white;
text-align: center;
padding: 25px;
}
.container {
width: 90%;
max-width: 800px;
margin: 25px auto;
}
section {
background: white;
padding: 20px;
margin-bottom: 20px;
border-radius: 8px;
box-shadow: 0 3px 8px #ccc;
}
h2 {
color: #174a7c;
}
.url {
background: #222;
color: white;
padding: 12px;
border-radius: 5px;
}

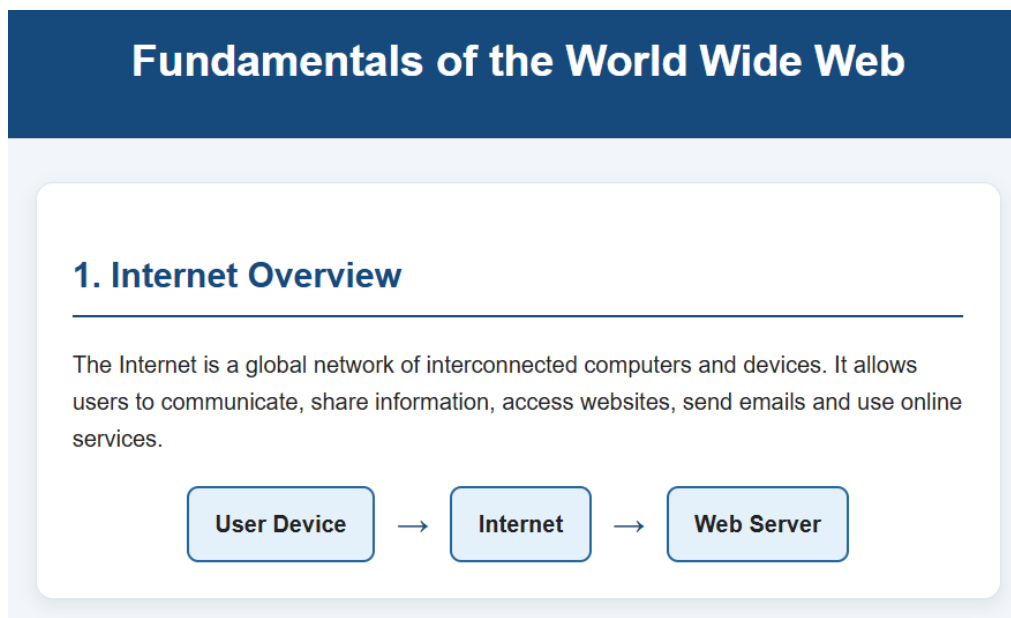
```

```

}
.diagram {
display: flex;
justify-content: center;
align-items: center;
gap: 15px;
}
.diagram div {
background: #e3f0fa;
border: 2px solid #174a7c;
padding: 12px;
border-radius: 5px;
}
table {
width: 100%;
border-collapse: collapse;
}
th, td {
border: 1px solid #ccc;
padding: 10px;
}
th {
background: #174a7c;
color: white;
}

```

Output:



2. Fundamental Computer Network Concepts

A computer network is a collection of connected devices that communicate and share resources.

IP Address

Identifies a device on a network.

Router

Connects different networks.

Protocol

Rules used for communication.

DNS

Converts domain names into IP addresses.



3. URL and its Structure

URL stands for **Uniform Resource Locator**. It is the address used to locate a resource on the Internet.

```
https://www.example.com/index.html
```

https://

Protocol

www

Subdomain

example.com

Domain Name

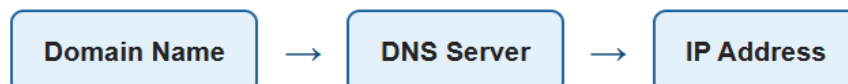
/index.html

Path

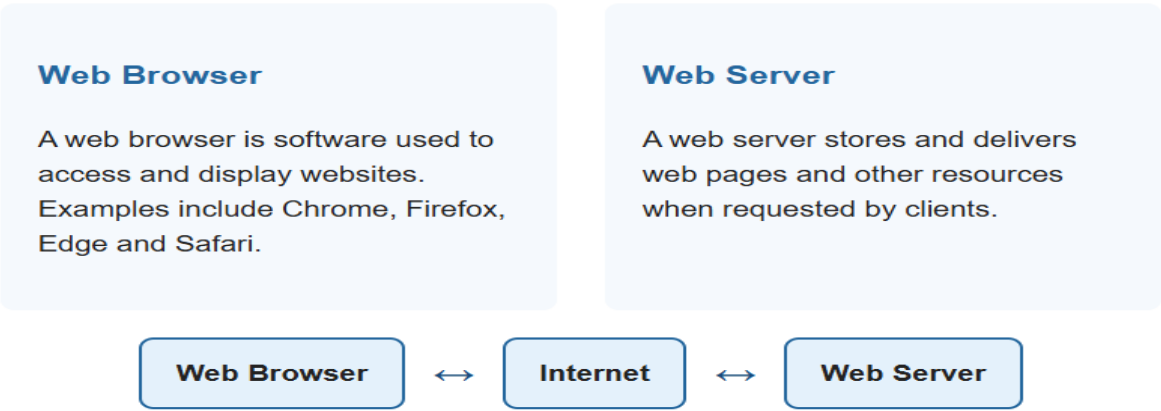
4. Domain Name

A domain name is a human-readable name used to identify a website on the Internet. It is easier to remember than an IP address.

```
www.example.com
```



5. Web Browsers and Web Servers



6. Working Principle of a Website

- 1. User enters the URL in the browser.
- 2. Browser contacts the DNS server.
- 3. DNS finds the IP address of the server.
- 4. Browser sends a request to the web server.
- 5. Server processes the request.
- 6. Server sends the required files to the browser.
- 7. Browser interprets HTML, CSS and JavaScript.
- 8. The webpage is displayed to the user.



7. Client-side vs Server-side Scripting

Client-side Scripting	Server-side Scripting
Runs in the user's browser.	Runs on the web server.
JavaScript is commonly used.	PHP, Python, Java and Node.js can be used.
Used for user interface interactions.	Used for business logic and data processing.
Example: Form validation.	Example: Database processing.

Explanation of Code:

- The <header> displays the main title of the webpage.
- The <section> elements divide the webpage into different topics such as the Internet, URL, browser, server, and scripting.
- The <div class="url"> displays an example URL and its different parts.
- The .diagram section shows the communication between the browser, Internet, and server.
- The <table> is used to compare client-side and server-side scripting.
- CSS is used to add background colors, borders, spacing, shadows, and text colors.
- display: flex is used to arrange the browser, Internet, and server boxes in a row.

Conclusion:

Thus, a simple webpage explaining the fundamentals of the World Wide Web was successfully created using HTML and CSS. The webpage presents important concepts such as the Internet, URL structure, browser-server communication, and client-side and server-side scripting in a clear and organized manner.

2. **Scenario:** A college wants to digitize its student admission process.

Task: Design a responsive Student Admission Form using HTML5 and CSS3. Your webpage should

demonstrate:

- HTML Form Elements
- Different Input Types
- Media Elements (Image/Video/Audio)
- CSS3 Selectors
- Box Model
- Responsive User Interface

Program:

Index.html:

```
<!DOCTYPE html>  
<html>
```



```

<head>
<title>Student Admission Form</title>
<link rel="stylesheet" href="style.css">
</head>
<body>
<header>
<h1>Student Admission Form</h1>
<p>College Admission Portal</p>
</header>
<div class="container">

<form>
<h2>Personal Information</h2>
<label>Student Name</label>
<input type="text" placeholder="Enter name">
<label>Email</label>
<input type="email" placeholder="Enter email">
<label>Mobile Number</label>
<input type="tel" placeholder="Enter mobile number">
<label>Date of Birth</label>
<input type="date">
<label>Gender</label>
<div class="options">
<label><input type="radio" name="gender"> Male</label>
<label><input type="radio" name="gender"> Female</label>
</div>
<h2>Academic Information</h2>
<label>Course</label>
<select>
<option>Select Course</option>
<option>Computer Science</option>
<option>Electronics</option>
<option>Mechanical</option>
</select>
<label>Percentage</label>
<input type="number" min="0" max="100">
<label>Address</label>
<textarea rows="4"></textarea>
<label>Upload Photo</label>
<input type="file" accept="image/*">
<h2>Activities</h2>
<div class="options">
<label><input type="checkbox"> Sports</label>
<label><input type="checkbox"> Music</label>

```

```

<label><input type="checkbox"> Coding</label>
</div>
<button type="submit">Submit Application</button>
<button type="reset">Reset</button>
</form>
</div>
</body>
</html>

```

Style.css:

```

body {
font-family: Arial, sans-serif;
margin: 0;
background: #f2f6fa;
}
header {
background: #174a7c;
color: white;
text-align: center;
padding: 25px;
}
.container {
width: 90%;
max-width: 800px;
margin: 30px auto;
}
.container > img {
width: 100%;
height: 220px;
object-fit: cover;
border-radius: 10px;
}
form {
background: white;
padding: 30px;
margin-top: 20px;
border-radius: 10px;
box-shadow: 0 4px 12px #ccc;
}
h2 {
color: #174a7c;
border-bottom: 2px solid #174a7c;
padding-bottom: 7px;
}
label {
display: block;
font-weight: bold;
margin: 12px 0 6px;
}

```

```
input, select, textarea {
width: 100%;
padding: 10px;
border: 1px solid #bbb;
border-radius: 5px;
box-sizing: border-box;
}
.options {
display: flex;
gap: 20px;
}
.options label {
font-weight: normal;
}
.options input {
width: auto;
}
button {
padding: 12px 20px;
margin-top: 20px;
border: none;
border-radius: 5px;
background: #174a7c;
color: white;
cursor: pointer;
}
button[type="reset"] {
background: #777;
}
button:hover {
opacity: 0.8;
}
@media (max-width: 600px) {
.container {
width: 94%;
}
form {
padding: 20px;
}
.options {
flex-direction: column;
gap: 5px;
}
}
```

Output:

Student Admission Form

College Admission Portal



Personal Information

Student Name

Email

Mobile Number

Date of Birth
 

Gender
☐ Male ☒ Female

Academic Information

Course

Computer Science



Percentage

95.4

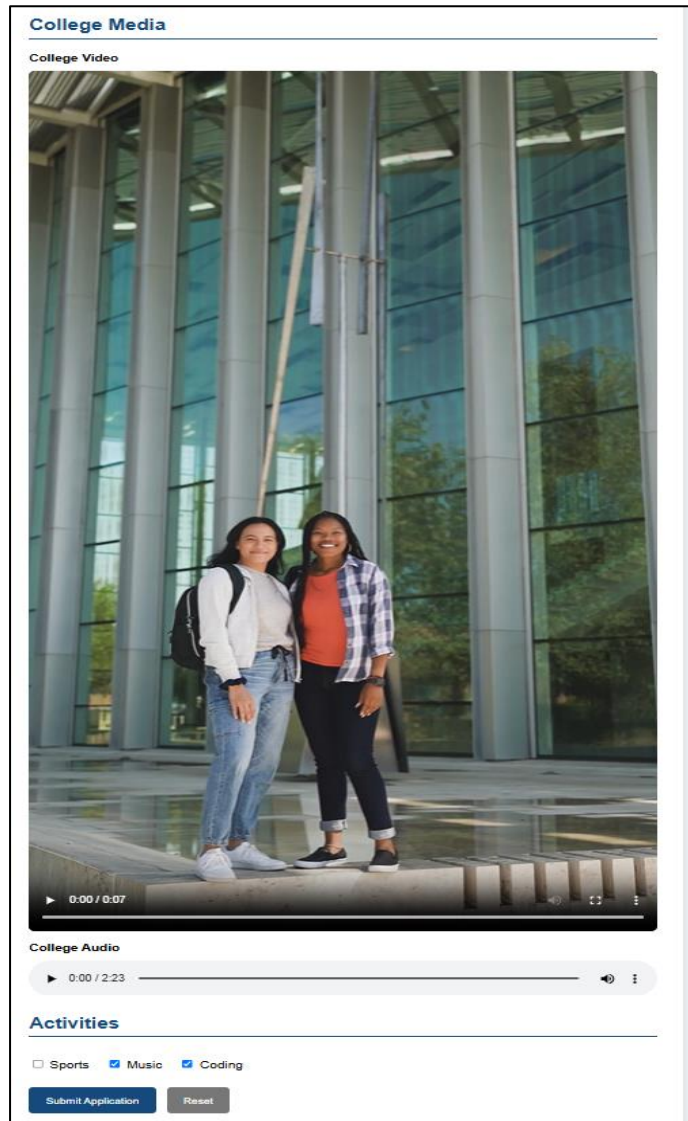
Address

Avinashi road,coimbatore.



Upload Photo

No file chosen



Explanation of Code:

- The <form> element is used to create the Student Admission Form.
- Different HTML5 input types such as text, email, tel, date, number, and file are used to collect student information.
- Radio buttons are used for selecting gender.
- Checkboxes are used for selecting multiple activities.
- The <select> element is used to select a course.
- <textarea> is used to enter the student's address.
- The element displays a college image.
- CSS is used to add colors, borders, padding, margins, and shadows.
- The :hover selector changes the button appearance when the mouse is placed over it.

- The @media rule makes the admission form responsive on smaller screens.

Conclusion:

Thus, a responsive Student Admission Form was successfully created using HTML5 and CSS3. The form includes different input types, selection controls, an image, and proper CSS styling. The responsive design allows the form to adjust according to different screen sizes.

3. **Scenario:** An online fashion store wants an attractive home page to improve customer engagement.

Task: Design the homepage using HTML5 and CSS3 by incorporating:

- Backgrounds and Borders
- Text Effects
- CSS Animations
- Multiple Column Layout
- User Interface Design Principles

Program:

Index.html

```
<!DOCTYPE html>
<html>
<head>
<title>StyleHub Fashion Store</title>
<link rel="stylesheet" href="style.css">
</head>
<body>
<header>
<h1>StyleHub</h1>
<nav>
<a href="#">Home</a>
<a href="#">Men</a>
<a href="#">Women</a>
<a href="#">Offers</a>
</nav>
</header>
<section class="hero">
```

```

<div>
<p>NEW COLLECTION 2026</p>
<h2>Wear Your Style</h2>
<p>Discover the latest fashion trends at affordable prices.</p>
<button>Shop Now</button>
</div>

</section>
<section class="products">
<h2>Featured Collections</h2>
<div class="product-grid">
<div class="product">

<h3>Summer Dress</h3>
<p>₹1,299</p>
<button>Buy Now</button>
</div>
<div class="product">

<h3>Casual Shirt</h3>
<p>₹899</p>
<button>Buy Now</button>
</div>
<div class="product">

<h3>Sports Shoes</h3>
<p>₹1,799</p>
<button>Buy Now</button>
</div>
</div>
</div>
<section class="offer">
<h2>Flat 30% OFF</h2>
<p>Limited time offer on selected products!</p>
<button>Shop Offers</button>
</section>
<footer>
<h3>StyleHub</h3>
<p>Fashion for every style.</p>
</footer>
</body>
</html>

```

Style.css:

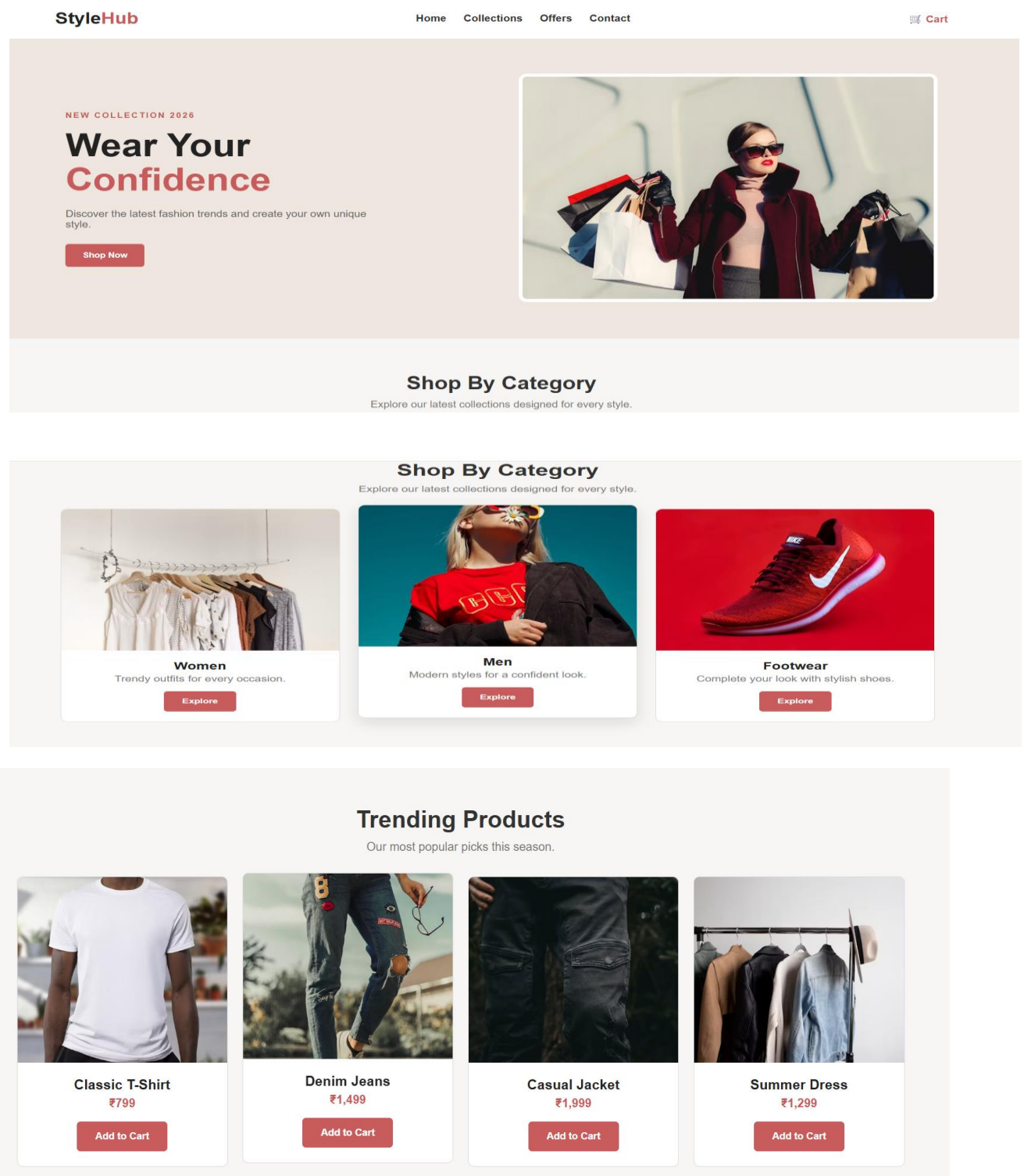
```
body {
margin: 0;
font-family: Arial, sans-serif;
background: #f8f5f2;
color: #222;
}
header {
background: #222;
color: white;
padding: 20px 8%;
display: flex;
justify-content: space-between;
align-items: center;
}
header h1 {
color: #e58b8b;
}
nav {
display: flex;
gap: 20px;
}
nav a {
color: white;
text-decoration: none;
}
nav a:hover {
color: #e58b8b;
}
.hero {
padding: 60px 8%;
display: grid;
grid-template-columns: 1fr 1fr;
gap: 40px;
align-items: center;
background: #eade5;
}
.hero h2 {
font-size: 45px;
color: #b74f4f;
text-shadow: 2px 2px #ddd;
}
.hero img {
width: 100%;
```



```
height: 350px;
object-fit: cover;
border: 5px solid white;
border-radius: 10px;
animation: slide 1.5s;
}
@keyframes slide {
from {
opacity: 0;
transform: translateX(40px);
}
to {
opacity: 1;
transform: translateX(0);
}
}
button {
background: #b74f4f;
color: white;
padding: 12px 22px;
border: none;
border-radius: 5px;
cursor: pointer;
}
button:hover {
background: #8f3838;
}
.products {
width: 85%;
margin: 50px auto;
text-align: center;
}
.product-grid {
display: grid;
grid-template-columns: repeat(3, 1fr);
gap: 25px;
}
.product {
background: white;
padding: 15px;
border: 1px solid #ddd;
border-radius: 10px;
}
.product:hover {
```

```
transform: translateY(-5px);
}
.product img {
width: 100%;
height: 250px;
object-fit: cover;
}
.product p {
color: #b74f4f;
font-weight: bold;
}
.offer {
text-align: center;
background: #b74f4f;
color: white;
padding: 50px;
}
.offer h2 {
font-size: 40px;
animation: pulse 2s infinite;
}
@keyframes pulse {
50% {
transform: scale(1.1);
}
}
footer {
background: #222;
color: white;
text-align: center;
padding: 25px;
}
@media (max-width: 700px) {
header {
flex-direction: column;
}
.hero {
grid-template-columns: 1fr;
text-align: center;
}
.product-grid {
grid-template-columns: 1fr;
}
```

Output:



Explanation of Code:

- The <header> contains the **StyleHub logo and navigation menu**.
- The .hero section introduces the fashion collection using text, an image, and a **Shop Now** button.

- CSS Grid is used to create a **multiple-column layout** for the products.
- Product cards contain images, product names, prices, and buttons.
- background, border, and border-radius properties are used to improve the visual appearance.
- text-shadow is used to create a **text effect** for the main heading.
- @keyframes slide creates an animation for the hero image.
- @keyframes pulse creates an animation for the special offer.
- The :hover selector changes the appearance of navigation links and buttons.
- The @media rule makes the webpage responsive on smaller screens.

Conclusion:

Thus, an attractive online fashion store homepage was successfully designed using HTML5 and CSS3. The webpage demonstrates backgrounds, borders, text effects, CSS animations, and a multiple-column product layout. Responsive design was also included to provide a better user experience on different screen sizes.

4. **Scenario:** A school wants to automate the student grade calculation process.

Task: Develop a JavaScript program that accepts marks of five subjects and displays the Total, Average, Grade, and Pass/Fail status. The solution should demonstrate:

- Variables and Data Types
- Literals
- Operators
- Statements (Selection and Iteration)
- User-defined Functions

Program:

```
<!DOCTYPE html>
<html>
<head>
<title>Student Grade Calculator</title>
<link rel="stylesheet" href="style.css">
</head>
<body>
<div class="container">
<h1>Student Grade Calculator</h1>
<input type="text" id="name" placeholder="Enter student name">
<input type="number" id="m1" placeholder="Subject 1">
<input type="number" id="m2" placeholder="Subject 2">
<input type="number" id="m3" placeholder="Subject 3">
<input type="number" id="m4" placeholder="Subject 4">
```

```

<input type="number" id="m5" placeholder="Subject 5">
<button onclick="calculate()">Calculate Result</button>
<div class="result">
<h2>Result</h2>
<p>Name: <b id="resultName">-</b></p>
<p>Total: <b id="total">-</b></p>
<p>Average: <b id="average">-</b></p>
<p>Grade: <b id="grade">-</b></p>
<p>Status: <b id="status">-</b></p>
</div>
</div>
<script src="script.js"></script>
</body>
</html>

```

Style.css:

```

body {
font-family: Arial, sans-serif;
background: #eef3f8;
}
.container {
width: 90%;
max-width: 500px;
margin: 40px auto;
background: white;
padding: 30px;
border-radius: 10px;
box-shadow: 0 4px 12px #ccc;
}
h1 {
text-align: center;
color: #174a7c;
}
input {
width: 100%;
padding: 12px;
margin: 7px 0;
box-sizing: border-box;
}
button {
width: 100%;
padding: 12px;
margin-top: 10px;

```

```
background: #174a7c;
color: white;
border: none;
cursor: pointer;
}
.result {
background: #eef5fb;
margin-top: 25px;
padding: 20px;
border-radius: 8px;
}
.result h2 {
color: #174a7c;
}
```

script.js:

```
function calculateTotal(marks) {
let total = 0;
for (let i = 0; i < marks.length; i++) {
total += marks[i];
}
return total;
}
```

```
function calculateGrade(average) {
if (average >= 90)
return "A+";
else if (average >= 80)
return "A";
else if (average >= 70)
return "B";
else if (average >= 60)
return "C";
else if (average >= 50)
return "D";
else
return "F";
}
```

```
function calculate() {
let name = document.getElementById("name").value;
let marks = [
Number(document.getElementById("m1").value),
Number(document.getElementById("m2").value),
```

```

Number(document.getElementById("m3").value),
Number(document.getElementById("m4").value),
Number(document.getElementById("m5").value)
];

let total = calculateTotal(marks);
let average = total / 5;
let grade = calculateGrade(average);
let status = "Pass";

for (let i = 0; i < marks.length; i++) {
  if (marks[i] < 40) {
    status = "Fail";
  }
}

document.getElementById("resultName").innerHTML = name;
document.getElementById("total").innerHTML = total;
document.getElementById("average").innerHTML = average.toFixed(2);
document.getElementById("grade").innerHTML = grade;
document.getElementById("status").innerHTML = status;
}

```

Output:

Student Grade Calculator

Enter student details and marks

Student Name

Subject 1

Subject 2

Subject 3

Subject 4

Subject 5

Calculate Result

Reset

Student Result

Student Name SRIMATHI K	
Total 478	Average 95.60
Grade A+	Status Pass

Explanation of Code:

- The HTML page accepts the **student name and marks for five subjects**.
- The calculateTotal() function uses a for loop to calculate the total marks.
- The average is calculated by dividing the total by 5.
- The calculateGrade() function uses if-else statements to determine the grade.
- A for loop checks whether any subject mark is below 40 and determines the **Pass/Fail status**.
- Number() converts the entered marks into numeric values.
- getElementById() is used to display the calculated result on the webpage.
- CSS is used to style the calculator with a simple form, colors, spacing, and result box.

Conclusion:

Thus, a **Student Grade Calculator** was successfully developed using HTML, CSS, and JavaScript. The program accepts the student's name and five subject marks and calculates the total, average, grade, and pass/fail status. It demonstrates JavaScript variables, operators, conditional statements, loops, arrays, and user-defined functions.

5. **Scenario:** A supermarket wants to generate customer bills automatically.

Task: Develop a JavaScript program to calculate the bill amount for three purchased products. The

program should:

- Accept quantity and unit price for each product
- Calculate the total bill
- Apply a 10% discount if the total amount exceeds ₹2000

- Display the final payable amount

The solution should demonstrate:

- Variables and Data Types
- Literals
- Operators
- Statements
- User-defined Functions

Program:

Index.html:

```
<!DOCTYPE html>

<html>

<head>

<title>Supermarket Billing</title>

<link rel="stylesheet" href="style.css">

</head>

<body>

<div class="container">

<h1>Supermarket Billing System</h1>

<input type="text" id="p1" placeholder="Product 1">

<input type="number" id="q1"

placeholder="Quantity">

<input type="number" id="pr1" placeholder="Unit

Price">

<input type="text" id="p2" placeholder="Product 2">

<input type="number" id="q2"
```

```

placeholder="Quantity">

<input type="number" id="pr2" placeholder="Unit
Price">

<input type="text" id="p3" placeholder="Product 3">

<input type="number" id="q3"
placeholder="Quantity">

<input type="number" id="pr3" placeholder="Unit
Price">

<button onclick="generateBill()">Generate
Bill</button>

<div class="result">

<h2>Bill Summary</h2>

<p id="item1">Product 1: -</p>

<p id="item2">Product 2: -</p>

<p id="item3">Product 3: -</p>

<hr>

<p>Total: ₹<span id="total">0</span></p>

<p>Discount: ₹<span id="discount">0</span></p>

<h3>Final Amount: ₹<span id="final">0</span></h3>

</div>

</div>

<script src="script.js"></script>

</body>

</html>

```

Style.css:

```
body {  
  
margin: 0;  
  
font-family: Arial, sans-serif;  
  
background: #f1f4f6;  
  
}  
  
.container {  
  
width: 90%;  
  
max-width: 600px;  
  
margin: 40px auto;  
  
background: white;  
  
padding: 30px;  
  
border-radius: 10px;  
  
box-shadow: 0 4px 12px #ccc;  
  
}  
  
h1 {  
  
text-align: center;  
  
color: #174a7c;  
  
}  
  
input {  
  
width: 100%;  
  
padding: 11px;  
  
margin: 6px 0;  
  
box-sizing: border-box;  
  
}  
  
button {
```

```
width: 100%;  
padding: 12px;  
margin-top: 15px;  
background: #174a7c;  
color: white;  
border: none;  
cursor: pointer;  
}
```

```
.result {  
margin-top: 25px;  
background: #f0f6fa;  
padding: 20px;  
border-radius: 8px;  
}
```

```
.result h2 {  
color: #174a7c;  
}
```

```
.result h3 {  
background: #dcecf7;  
padding: 12px;  
color: #16833b;  
}
```

Script.js:

```
function calculateAmount(quantity, price) {  
return quantity * price;
```

```
}

function calculateDiscount(total) {

if (total > 2000) {

return total * 0.10;

}

return 0;

}

function generateBill() {

let product1 = document.getElementById("p1").value;

let product2 = document.getElementById("p2").value;

let product3 = document.getElementById("p3").value;

let quantity1 =

Number(document.getElementById("q1").value);

let quantity2 =

Number(document.getElementById("q2").value);

let quantity3 =

Number(document.getElementById("q3").value);

let price1 =

Number(document.getElementById("pr1").value);

let price2 =

Number(document.getElementById("pr2").value);

let price3 =

Number(document.getElementById("pr3").value);

let amount1 = calculateAmount(quantity1, price1);

let amount2 = calculateAmount(quantity2, price2);
```

```

let amount3 = calculateAmount(quantity3, price3);

let total = amount1 + amount2 + amount3;

let discount = calculateDiscount(total);

let finalAmount = total - discount;

document.getElementById("item1").innerHTML =

product1 + " = ₹" + amount1;

document.getElementById("item2").innerHTML =

product2 + " = ₹" + amount2;

document.getElementById("item3").innerHTML =

product3 + " = ₹" + amount3;

document.getElementById("total").innerHTML =

total.toFixed(2);

document.getElementById("discount").innerHTML =

discount.toFixed(2);

document.getElementById("final").innerHTML =

finalAmount.toFixed(2);

}

```

Output:

The screenshot displays the 'QuickMart Supermarket Billing System' interface. At the top, there's a dark header with the QuickMart logo and a 'BILL' button. Below the header, a section titled 'Create Your Bill' prompts the user to 'Enter the details of three purchased products.' The main area contains three columns for product entry:

- Product One (01):** Product Name: rice, Quantity: 1, Unit Price (₹): 100.
- Product Two (02):** Product Name: oil, Quantity: 2, Unit Price (₹): 150.
- Product Three (03):** Product Name: soap, Quantity: 3, Unit Price (₹): 60.

At the bottom, there are two buttons: 'Generate Bill' (orange) and 'Clear' (grey).

Bill Summary

QuickMart • Customer Receipt



Product	Qty	Price	Amount
rice	1	₹100.00	₹100.00
oil	2	₹150.00	₹300.00
soap	3	₹60.00	₹180.00

Subtotal ₹580.00

Discount (10%) ₹0.00

Final Payable ₹580.00

Explanation of Code:

- The HTML page accepts the **name, quantity, and unit price of three products**.
- The calculateAmount() function calculates the price of each product using **quantity × unit price**.
- The three product amounts are added to calculate the **total bill**.
- The calculateDiscount() function checks whether the total is greater than **₹2000**.
- If the total is greater than ₹2000, a **10% discount** is calculated.
- The final payable amount is calculated by subtracting the discount from the total.
- Number() converts the entered values into numbers.
- getElementById() displays the calculated bill on the webpage.
- CSS is used to create a simple billing interface with a result section..

Conclusion:

Thus, a **Supermarket Billing System** was successfully developed using HTML, CSS, and JavaScript. The program calculates the total cost of three products and automatically applies a 10% discount when the total exceeds ₹2000. It demonstrates variables, arithmetic operators, conditional statements, functions, and DOM manipulation.